

Knowing the Outer Function vs. Observing the Intermediates

Splitting the composite-BO gain into its two parts

`github.com/MuhammadAli-CS/three-arm-composite`

Where we are

Composite problems: $f(x) = g(h(x))$, with h an expensive black box and g a known, cheap function.

- **Direct**: fit a GP to f , ignore the structure.
- **Composite**: fit GPs to h , push samples through g .

Composite won most benchmarks — by anywhere from **121%** to **nothing**.

The question this work answers

We do not know *why* it wins, because composite changes two things at once.

The confound

Going from direct to composite gives the surrogate **two** advantages simultaneously:

1. **It knows** g — handed the outer function, never learns it from data.
2. **It observes** h — each expensive run returns k numbers, not one.

Every result we have bundles these together.

- Where composite wins big — which advantage did it?
- Where composite wins nothing — which one failed?

Two numbers cannot separate two effects.

The fix: a third arm

	g unknown	g known
observes y only	arm 1 direct	arm 3 latent
observes h	— (incoherent)	arm 2 composite

Arm 3 knows g exactly, holds the *same* GP prior on h as composite, but never sees h — it must infer it from y .

latent — direct = the value of **knowing** g
composite — latent = the value of **observing** h

The two sum to the composite gain we already publish — so the decomposition must add up, and any failure is a bug, not a finding.

What should govern the split

Observing h can only help where $y = g(h)$ fails to pin h down. **Invertibility of g is the controlling property.**

g injective

$h = g^{-1}(y)$, so arms 2 and 3 see identical information.

$\Rightarrow$ observing h worth **exactly 0**

g linear

A sum of independent GPs *is* a GP, so arm 3 collapses to arm 1.

$\Rightarrow$ knowing g worth ≈ 0

Why the linear case matters for the paper

Truss and RCM40 both have linear outer maps — and both gave our weakest results.

In between: what g destroys is the *fiber* $g^{-1}(y)$ — two points for z^2 , several for $\cos$, a half-line for a clipped map, a circle for a norm.

How arm 3 infers h — the whole idea in one example

$g(z) = z^2$. You observe $y = 4$ at **two nearby inputs**.

Four candidates fit the data *perfectly*: $(+2, +2)$, $(-2, -2)$, $(+2, -2)$, $(-2, +2)$.

The data cannot choose. **Smoothness can**: a jump from $+2$ to -2 over a tiny step is not a plausible function.

distance between inputs	correlation	same-sign advantage
close	0.9	2.9×10^{16}
medium	0.3	$14 \times$
far	0.02	$1.2 \times$

Note what is learned: *not* whether $h = +2$ or -2 (still 50/50), but that both points did the **same** thing.

That is Bayes' rule

Score every candidate filling-in of h on two questions:

$$p(\mathbf{h} \mid \mathbf{y}) \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i; g(h(x_i)), \sigma^2)}_{\text{does it match the data?}} \cdot \underbrace{\prod_{j=1}^k \mathcal{N}(h_j(X); m_j, v_j K)}_{\text{is it a plausible function?}}$$

Neither factor works alone: the likelihood cannot choose between ± 2 ; the prior does not know the data.

Why this is the hard part

Arms 1 and 2 are ordinary GP regression — closed form. A Gaussian prior times a *non*-Gaussian likelihood is not Gaussian, so arm 3 has **no formula** and needs MCMC. Every difficulty in this project has been here.

What the two factors actually compute

First, the notation. $\mathcal{N}(a; b, c)$ is not a distribution — it is a **number**: the height of the bell curve with mean b and variance c , read off at the position a . Three numbers in, one number out.

Factor 1 — does it match?

Centre the bell on what the candidate *predicts*, $g(h(x_i))$; read it off at what you *measured*, y_i .

σ is the tolerance, and it sits in the denominator of the exponent — at $\sigma = 0.01$, missing by 0.1 costs e^{-50} .

$\prod_i$ runs over evaluated points: once $\mathbf{h}$ is fixed, the measurements are independent.

Factor 2 — is it smooth?

One multivariate Gaussian over the *whole vector* of values, with

$$K_{ii'} = \exp(-\|x_i - x_{i'}\|^2 / 2\ell^2).$$

Three points at $x = (0, 0.1, 0.6)$, $\ell = 0.5$:

$$K = \begin{pmatrix} 1 & 0.98 & 0.49 \\ 0.98 & 1 & 0.61 \\ 0.49 & 0.61 & 1 \end{pmatrix}$$

K^{-1} in that exponent **is** the smoothness machinery.

Evaluating it on three candidates

$g(z) = z^2$, $\sigma = 0.01$, measured $\mathbf{y} = (4, 4.2025, 1)$.

Candidate	$\mathbf{h}$	log-likelihood	log-prior	log-posterior
A smooth, matches	(2, 2.05, 1)	+11.06	-2.80	+8.26
B jumpy, matches	(2, - 2.05 , 1)	+11.06	-469.5	-458.45
C smooth, misses	(2, 2.02, 1.5)	-7876	-2.80	-7878.8

A and B have the identical log-likelihood — not similar, identical, because $(-2.05)^2 = (2.05)^2$. That factor is mathematically incapable of separating them. The prior does it alone: $\mathbf{h}^\top K^{-1} \mathbf{h}$ is 4.42 for A and 937.8 for B, because $K_{12} = 0.98$ demands that the first two points agree.

C inverts it — a marginally *better* prior than A, destroyed by predicting 2.25 where you measured 1.

Each candidate is killed by a different factor. That is why the equation is a product.

The experiment

19 outer maps applied to inner functions **drawn from the same GP family the surrogates assume**, so h is never a confound.

Group	Maps	Purpose
Controls	$z, e^z, \sinh z, y_1 + y_2$	answers known in advance
Sweep	$(z - c)^2, c = 0, 0.5, 1, 2, 3$	one knob on <i>how much</i> is destroyed
Pairs	$ z , \cos(\omega z),$ censoring maps	one property changed at a time
Analogs	sum, product, norm, max, ratio	mirror Truss, DTLZ2, Color, Lake, RGB

$d = 6$, 5 initial points + 40 evaluations, **4 function draws** $\times$ **5 designs**, all arms sharing the initial design and acquisition.

$\Rightarrow$ 380 BO runs per arm, $\approx$ 40 min on 16 cores.

All 19 conditions

Map	g	What g destroys		U_{lost}	obs. h	know g
Controls — answers known before the run						
identity	z	nothing		0.000	0.000*	+0.41
exp.b1	e^z	nothing (monotone)		0.000	0.000*	+0.63
sinh.b1	$\sinh z$	nothing (monotone)		0.000	0.000*	+0.44
sum	$y_1 + y_2$	how the sum splits	linear	0.702	-0.18	+0.12
The sweep — one form, loss tuned by c ; U_{lost} at iteration 0 in brackets						
shiftsq-c0	z^2	sign of z	[1.03]	0.280	+1.49	+0.12
shiftsq-c0.5	$(z - 0.5)^2$	sign of $z - 0.5$		0.283	+1.28	+0.63
shiftsq-c1	$(z - 1)^2$	sign of $z - 1$	[0.87]	0.283	+1.89	-0.10
shiftsq-c2	$(z - 2)^2$	rarely ambiguous	[0.10]	0.189	+2.75	-0.73
shiftsq-c3	$(z - 3)^2$	almost nothing	[0.01]	0.232	+2.51	-1.18
Matched pairs — one property changed at a time						
abs	$ z $	sign of z , with a kink		0.474	-0.06	-0.01
cos_w1	$\cos z$	which of few preimages		2.434	+2.45	-0.79
cos_w2	$\cos 2z$	which of more preimages		1.284	+1.06	+0.70
cos_w4	$\cos 4z$	which of many preimages		1.161	+1.08	-0.80
relu	$-\max(z, 0)$	all $z < 0$, away from opt.		0.075	-0.07	+0.20
clip-c2	$\text{clip}(2y_1) + y_2$	clipped coord., at opt.		0.580	+1.11	+0.28
Benchmark analogs — sum above is the Truss / RCM40 analog						
product	$y_1 y_2$	factoring, and sign	DTLZ2	1.640	+0.21	-0.41
norm	$\sqrt{y_1^2 + y_2^2}$	direction	Color	0.426	-0.10	-0.02
max	$\max(y_1, y_2)$	the smaller value	Lake	0.358	+0.14	+0.07
ratio	$u_1 / (0.05 + u_1 + u_2)$	scale	RGB	0.267	+0.12	-0.42

Differences in final $\log_{10}$ regret, > 0 favours the arm that sees more. "obs. h " has the measured +0.41 bias subtracted; * exact, no correction needed. SEs

0.06–0.48.

Controls are not optional

The first full grid looked **completely reasonable** — smooth trends, sensible signs, tight error bars.

It was wrong.

identity: “observing h ” must be **exactly 0**.

The code said **$+0.956 \pm 0.274$** .

Three defects, all in arm 3’s sampler. The decisive one: new points were initialised *off* the constraint surface $\{g(h) = y\}$, which the sampler can never travel to once the likelihood is tight. Residual grew over iterations:

$0.0001 \rightarrow 0.004 \rightarrow 0.027 \rightarrow 0.019 \rightarrow 0.043 \rightarrow \mathbf{0.121}$

After the fix: **-0.096 ± 0.175** , $U_{\text{lost}} = 0$ exactly.

Results: three predictions held

Prediction	Condition	Measured
observing $h = 0$ exactly	$z, e^z, \sinh z$	-0.000 ± 0.000
knowing $g \approx 0$	$y_1 + y_2$ (linear)	$+0.117 \pm 0.340$
censoring away from optimum ≈ 0	$-\max(z, 0)$	-0.066 ± 0.209
censoring at the optimum > 0	clipped pair	$+1.108 \pm 0.421$

The linear row is the one that bears on the paper: **on Truss and RCM40 there is no knowledge to exploit**, so whatever composite gains there must come from the extra observations alone.

The finding: where, not how much

Both maps censor, and `relu` discards strictly more of the domain — an entire half-line collapses to one value. But *where* it discards it is different:

Map	Censoring sits	U_{lost}	observing h
$-\max(z, 0)$	away from the optimum	0.075	-0.066 ± 0.209
clipped pair	at the optimum	0.580	$+1.108 \pm 0.421$

The rest of the table agrees — sorting by U_{lost} does not sort by gain:

product destroys 1.64, pays +0.21 · sum destroys 0.70, pays -0.18 · abs destroys 0.47, pays -0.06

U_{lost} already encodes this

It is measured at the points BO *actually evaluated*, so a fiber that sits away from the optimum barely registers — 0.075 against 0.580.

Sharper hypothesis

The question for a new problem is not “is g invertible?” but **“is g invertible near the Pareto front?”**

What did not hold: the sweep

$(z - c)^2$ — same form, one parameter, information loss tuned by c . Designed to be the cleanest experiment in the set.

c	0	0.5	1	2	3
U_{lost} at iteration 0	1.026	—	0.869	0.104	0.013
observing h (predicted)	large		↓	↓	→ 0
observing h (measured)	+1.49	+1.28	+1.89	+2.75	+2.51

The knob *does* turn on the initial design. Then, over BO iterations, U_{lost} falls for small c and **ris**es for large c — more data cannot increase uncertainty about points already observed.

Second symptom: “knowing g ” goes **negative** (−1.18 at $c = 3$). Arm 3 holds the true kernel *and* knows g ; it should not lose to a plain GP.

Diagnosis so far

Hypothesis	Test	Outcome
Sampler budget too small	4 chains/1500 kept vs. 2/200	ruled out
Warm starts accumulate error	cold vs. warm, same design	ruled out
Level-set moves fail	acceptance in loop vs. in checks	confirmed

Move acceptance: 0.5–0.68 on spread designs, 0.000 on clustered ones. Flipping one point's sign against its packed neighbours is the 10^{16} -to-one rejection from the two-point example. The sampler stops changing its mind exactly when BO tightens the design.

Fix: block moves — flip a whole neighbourhood at once.

But that is not the whole story

identity with the sampler forced on gives “observing h ” = $+0.409 \pm 0.133$ with $U_{\text{lost}} = 0.0016$ — it *has* the information and still loses. Raising the MC budget $16\times$ only moves it to $+0.275$.

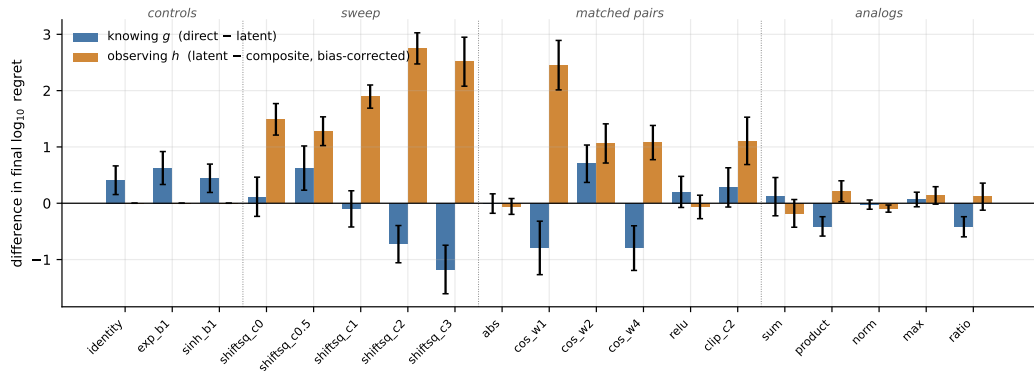
Analog vs. the benchmark results

Analog	Benchmark	observing h	Composite gain in the paper
product	DTLZ2	+0.213	largest (121%)
max	Lake	+0.139	moderate
ratio	RGB	+0.117	none
norm	Color	−0.095	moderate
sum	Truss, RCM40	−0.180	weakest

Ordering matches at both ends.

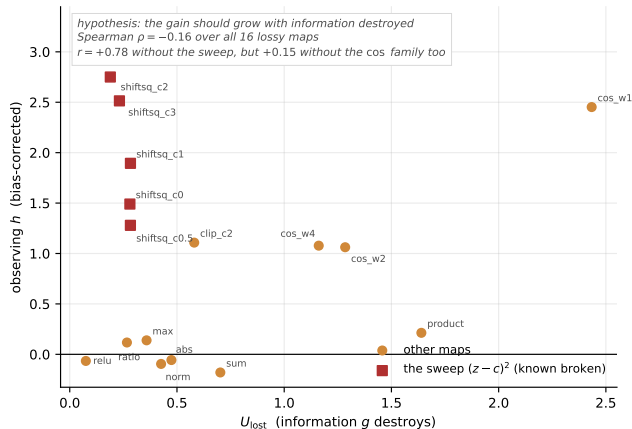
Caveat, stated plainly: standard errors are 0.06–0.25, so no individual pair separates significantly, and the correction subtracted from all of them is itself uncertain. Consistent with the benchmarks — not confirmation of them.

Both effects, every condition



Error bars are paired standard errors over the 20 (draw, seed) pairs. Note the sweep: *knowing g* turns *negative* as c grows, which cannot happen if arm 3 is working — it holds the true kernel *and* knows g .

The plot this study was built to produce



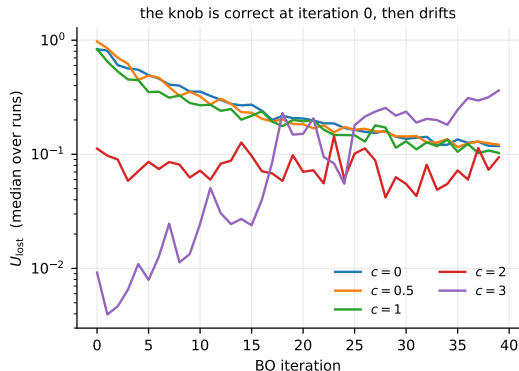
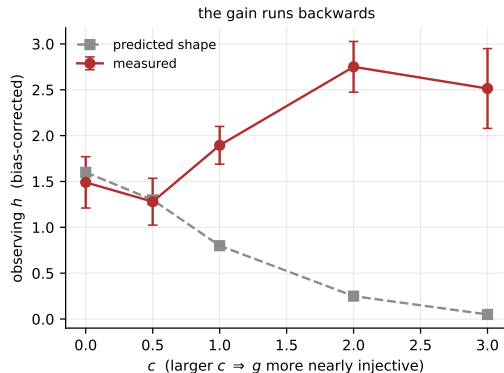
The hypothesis predicts an upward trend.

Over all 16 lossy maps there is none.

Dropping the sweep produces $r = +0.78$ — but dropping the cos family as well leaves $+0.15$, so the apparent trend rests on three points.

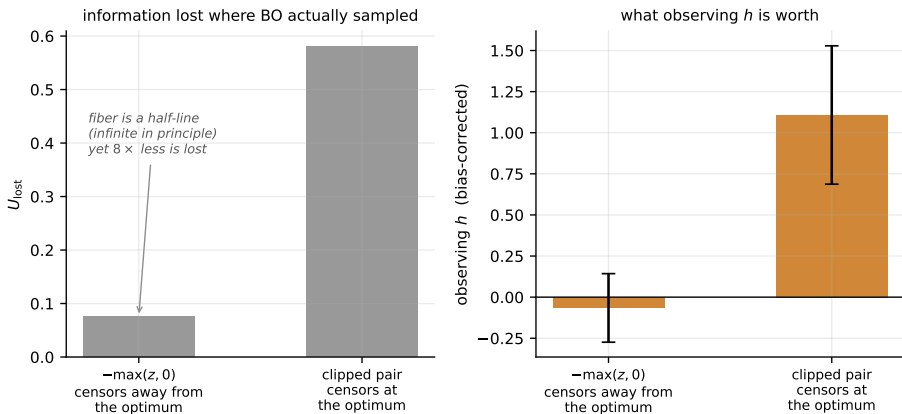
No line is fitted, deliberately: a regression here would assert a relationship the data do not support.

The sweep, and why it fails



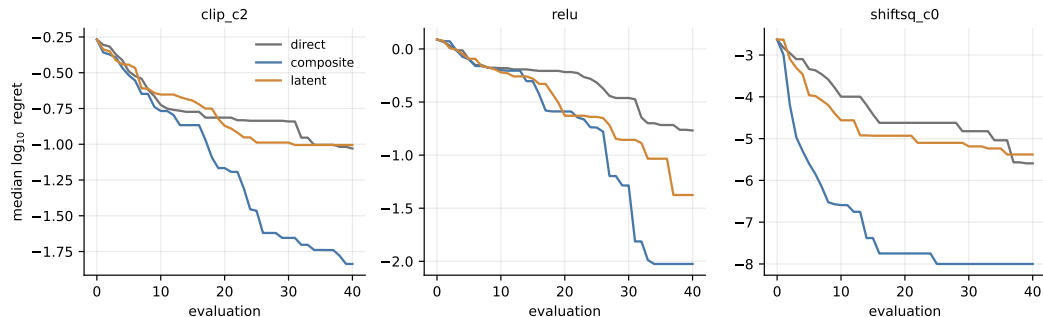
Right panel is the diagnosis: at $c = 3$ the latent arm starts knowing h almost exactly ($U_{\text{lost}} = 0.004$) and ends two orders of magnitude worse. Small c behaves correctly — uncertainty falls as data arrives. More data cannot increase uncertainty about points already observed.

The finding, in one figure



$-\max(z, 0)$ has the *larger* fiber — an entire half-line — but it sits away from the optimum, so little is lost where BO looks and observing h buys nothing. The clipped pair censors where the optimizer is working, and observing h is worth $13\times$ in final regret.

Convergence, three arms



Median over the 20 pooled runs, and **uncorrected** — so each latent–composite gap here still contains the +0.41 handicap that the identity control measures. On `relu` that handicap is most of the visible gap; on `clip_c2` and `shiftsq_c0` it is a small part of it.

Where this goes next

To fix, in order

1. The +0.275 residual on identity — smallest case that still shows the defect: one intermediate, no multimodality, near-point posterior. Everything else is downstream.
2. Block level-set moves.
3. All-MLE hyperparameters in every arm (removes the last asymmetry rather than correcting for it).

Cheapest test of the finding — no new BO runs

- Compute U_{lost} on each *real* benchmark restricted to near-Pareto-optimal points.
- It should predict the composite gain better than U_{lost} over the whole domain.
- Uses points every existing run already evaluated.

Summary

- Composite BO bundles **knowing** g with **observing** h ; a third arm separates them, and the three numbers must telescope.
- Arm 3 is the hard part: no closed form, so MCMC over quantities never observed.
- **Held**: observing h is exactly 0 when g is injective; knowing g is ≈ 0 when g is linear (Truss, RCM40).
- **Finding**: **where** g destroys information matters more than how much — censoring away from the optimum costs nothing, censoring at it costs $13\times$ in final regret.
- **Open**: the sweep runs backwards; a $+0.275$ bias survives on a control where the answer is provably 0.

Code, results and full write-up: github.com/MuhammadAli-CS/three-arm-composite